# ST7701S Application Circuit

**Preliminary Ver 0.1**
**2017/02**

**DISCLAIMER: THIS DOCUMENT IS PROVIDED TO YOU "AS IS" WITH NO WARRANTIES WHATSOEVER, INCLUDING ANY WARRANTY OF MERCHANTABILITY NON-INFRINGEMENT, OR FITNESS FOR ANY PARTICULAR PURPOSE. SITRONIX AND THE AUTHORS OF THIS DOCUMENT DISCLAIM ALL LIABILITY, INCLUDING LIABILITY FOR INFRINGEMENT OF ANY PROPRIETARY RIGHTS, RELATING TO USE OR IMPLEMENTATION OF INFORMATION IN THIS DOCUMENT. THE PROVISION OF THIS DOCUMENT TO YOU DOES NOT PROVIDE YOU WITH ANY LICENSE, EXPRESS OR IMPLIED, BY ESTOPPEL OR OTHERWISE, TO ANY INTELLECTUAL PROPERTY RIGHTS**

# LIST OF CONTENT


**1 HSD Panel ..... 4**

1.1 HSD 3.97 ((HSD040B8W9-A01) Application FPC Circuit ..... 4

1.1.1 MIPI Interface ..... 4

1.1.2 RGB Interface ..... 4

1.2 HSD 4.5” (HSD045B8W7-A00) Application FPC Circuit ..... 5

1.2.1 MIPI Interface ..... 5

1.2.2 RGB Interface ..... 5

1.3 HSD 5.0 (HSD050F8W4-A00) Application Circuit ..... 6

1.3.1 MIPI Interface ..... 6

1.3.2 RGB Interface ..... 6

**2 IVO Panel ..... 7**

2.1 IVO 3.97(C039SWP4) Application Circuit ..... 7

2.1.1 MIPI Interface ..... 7

2.1.2 RGB Interface ..... 7

2.2 IVO 4.5” TN (C045F8W7-C00)/ IPS (C045SWY2-0) Application Circuit ..... 8

2.2.1 MIPI Interface ..... 8

2.2.2 RGB Interface ..... 8

2.3 IVO 4.95 “(C050SWYG)/ 4.98” TN (C050SWY7-0) / IPS (C050SWY8-2) Application Circuit ……..9

2.3.1 MIPI Interface ..... 9

2.3.2 RGB Interface ..... 9

**3 CPT Panel ..... 10**

3.1 CPT 3.97(F040A16-604) Application Circuit ..... 10

3.1.1 MIPI Interface ..... 10

3.1.2 RGB Interface ..... 10

3.2 CPT 4.5(CLAG045LJ12) Application Circuit ..... 11

3.2.1 MIPI Interface ..... 11

3.2.2 RGB Interface ..... 11

**4 BOE Panel ..... 12**

4.1 BOE 3.97(B3 BD039WVS-810) Application Circuit ..... 12

4.1.1 MIPI Interface ..... 12

4.1.2 RGB Interface ..... 13

4.2 BOE 4.5 TN (BT045FWQ-N80)/ IPS (BF045WVQ-100) Application Circuit ..... 13

4.2.1 MIPI Interface ..... 13

4.2.2 RGB Interface ..... 14

4.3 BOE 5.0 TN (BT050FWQ-N80)/ IPS (BV050FWE-N43-1Q00) Application Circuit ..... 14

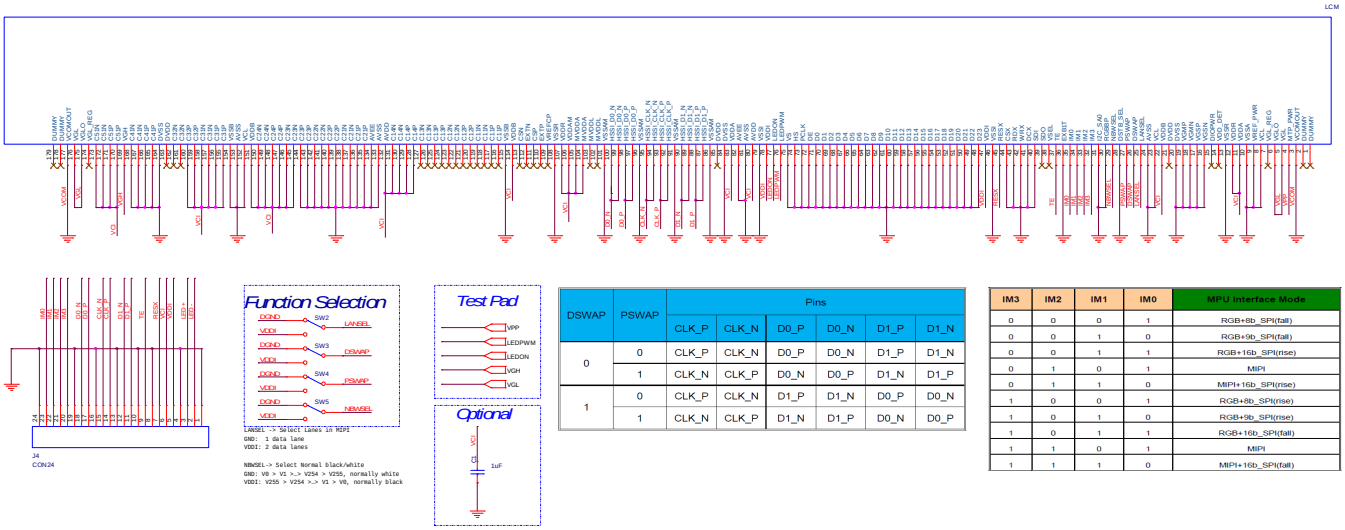
4.3.1 MIPI Interface ..... 14

4.3.2 RGB Interface ..... 15
**5 CMI Panel ..... 15**
5.1 CMI 5.5 Application Circuit ..... 15
5.1.1 MIPI Interface ..... 15
5.1.2 RGB Interface ..... 16
**6 CTC Panel ..... 16**
6.1 CTC 3.97 (PH040NA-01B) Application Circuit ..... 16
6.1.1 MIPI Interface ..... 16
6.1.2 RGB Interface ..... 16
6.2 CTC 4.5 () Application Circuit ..... 17
6.2.1 MIPI Interface ..... 17
6.2.2 RGB Interface ..... 17
**7 Support Panel Cell Model ..... 18**

# 1 HSD Panel

## 1.1 HSD 3.97 ((HSD040B8W9-A01) Application FPC Circuit

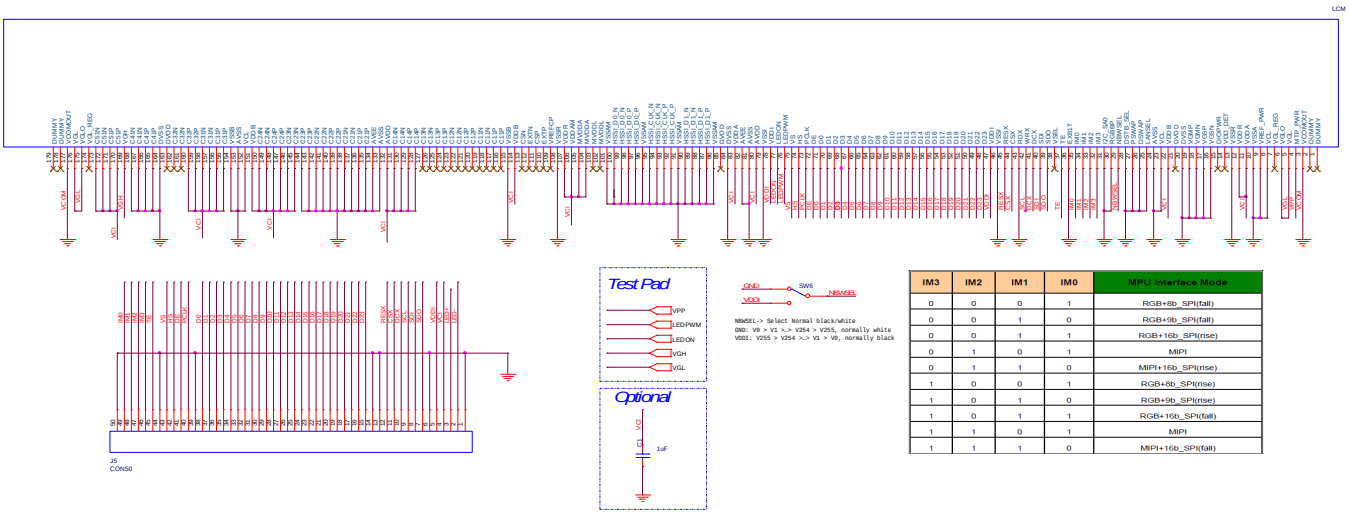
### 1.1.1 MIPI Interface




| DSWAP | PSWAP | Pins | | | | | |
|---|---|---|---|---|---|---|---|
| | | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| 0 | 0 | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| | 1 | CLK_N | CLK_P | D0_N | D0_P | D1_N | D1_P |
| 1 | 0 | CLK_P | CLK_N | D1_P | D1_N | D0_P | D0_N |
| | 1 | CLK_N | CLK_P | D1_N | D1_P | D0_N | D0_P |

| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(fall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(fall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(rise) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(fall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(fall) |

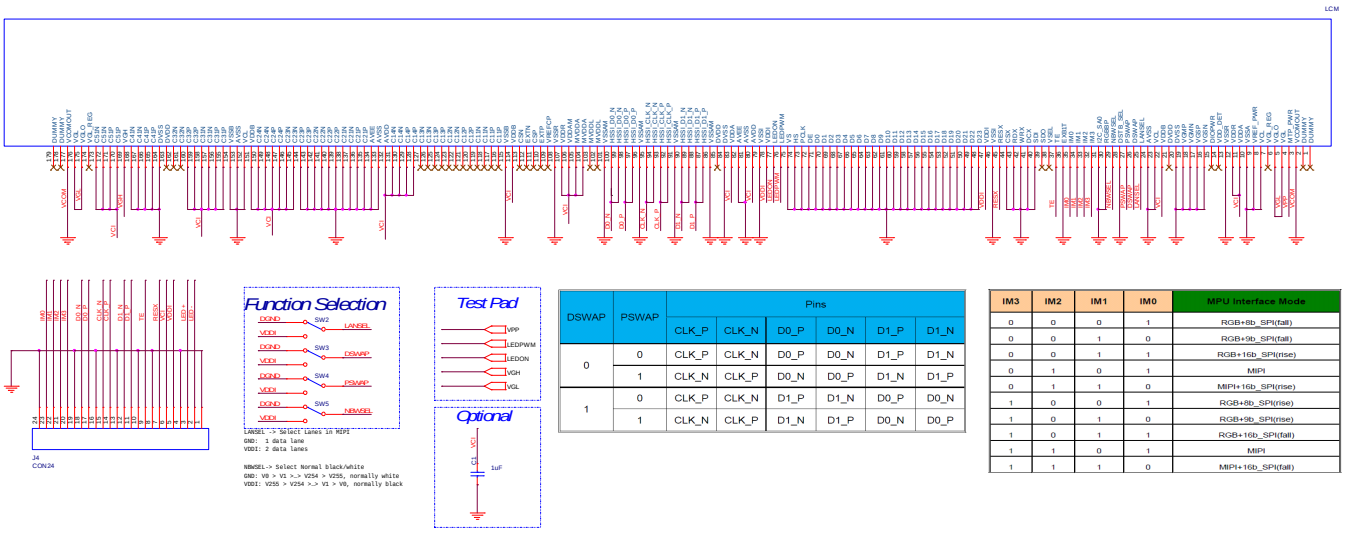
### 1.1.2 RGB Interface




| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(fall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(fall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(rise) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(fall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(fall) |

## 1.2 HSD 4.5” (HSD045B8W7-A00) Application FPC Circuit

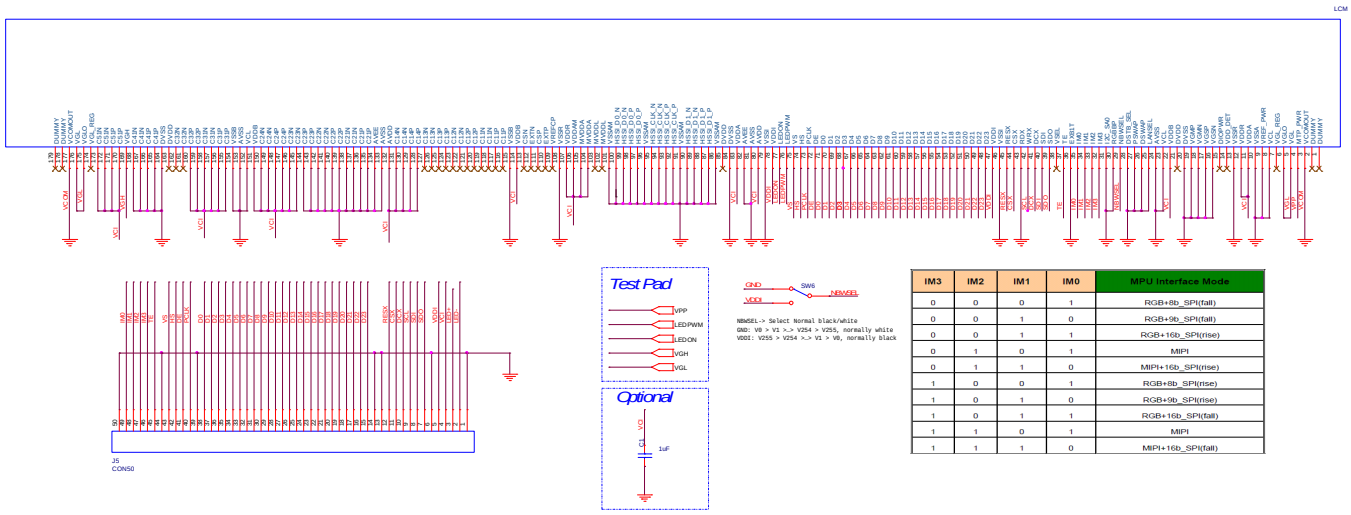
### 1.2.1 MIPI Interface




| DSWAP | PSWAP | Pins | | | | | |
|---|---|---|---|---|---|---|---|
| | | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| 0 | 0 | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| | 1 | CLK_N | CLK_P | D0_N | D0_P | D1_N | D1_P |
| 1 | 0 | CLK_P | CLK_N | D1_P | D1_N | D0_P | D0_N |
| | 1 | CLK_N | CLK_P | D1_N | D1_P | D0_N | D0_P |

| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(fall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(fall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(rise) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(fall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(fall) |

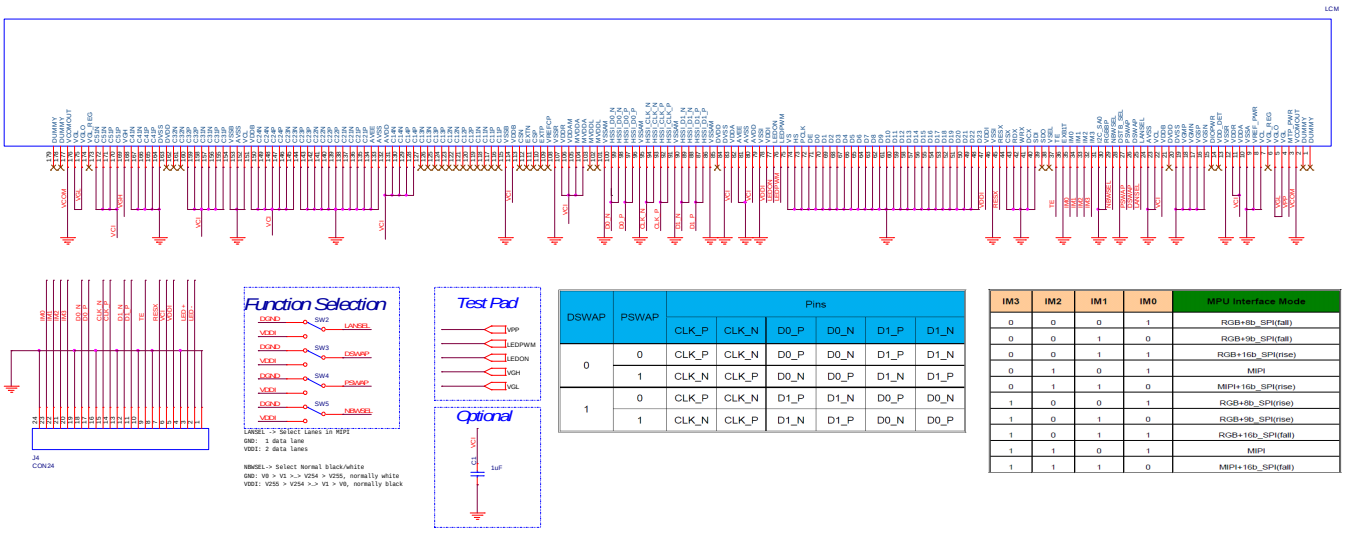
### 1.2.2 RGB Interface




| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(fall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(fall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(rise) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(fall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(fall) |

## 1.3 HSD 5.0 (HSD050F8W4-A00) Application Circuit

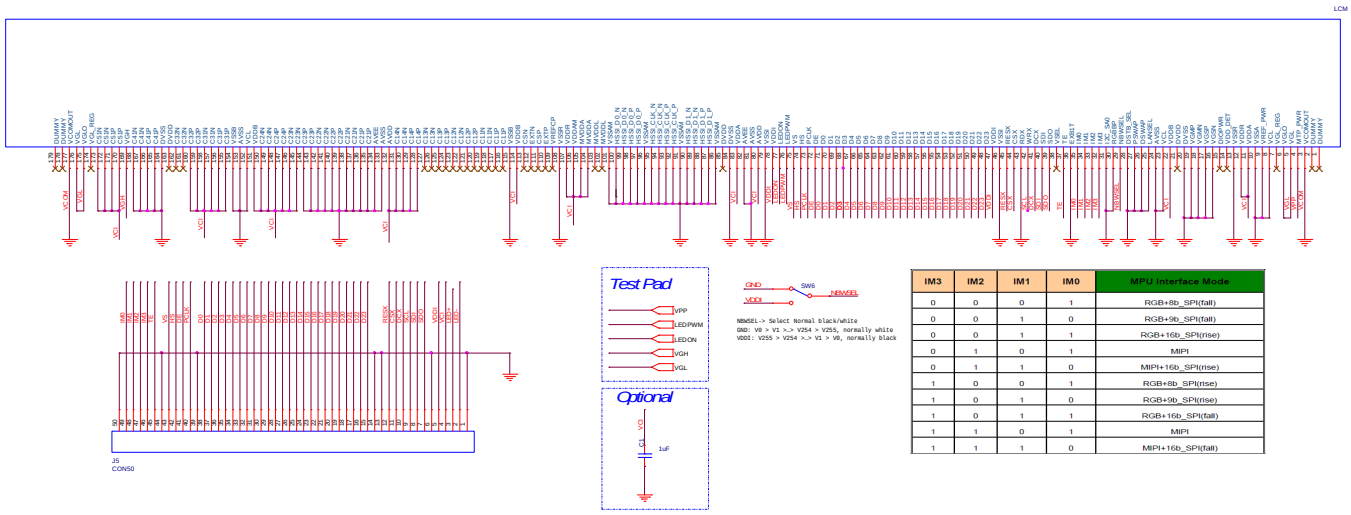
### 1.3.1 MIPI Interface




| DSWAP | PSWAP | Pins | | | | | |
|---|---|---|---|---|---|---|---|
| | | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| 0 | 0 | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| | 1 | CLK_N | CLK_P | D0_N | D0_P | D1_N | D1_P |
| 1 | 0 | CLK_P | CLK_N | D1_P | D1_N | D0_P | D0_N |
| | 1 | CLK_N | CLK_P | D1_N | D1_P | D0_N | D0_P |

| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(fall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(fall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(fall) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(fall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(fall) |

### 1.3.2 RGB Interface

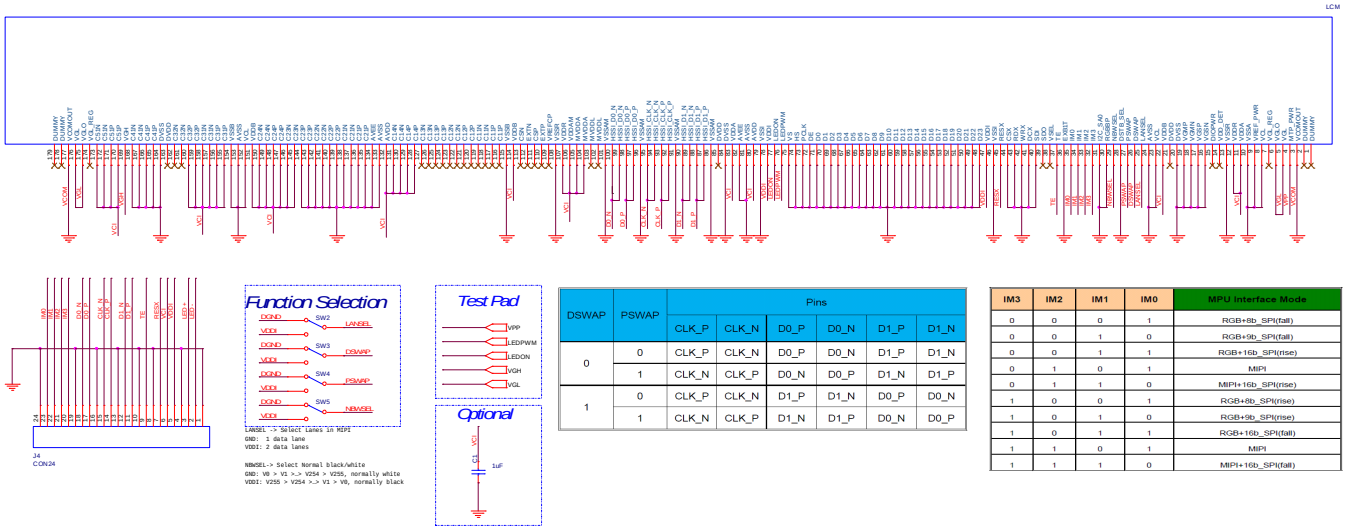



| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(fall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(fall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(rise) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(fall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(fall) |

# 2 IVO Panel

## 2.1 IVO 3.97(C039SWP4) Application Circuit

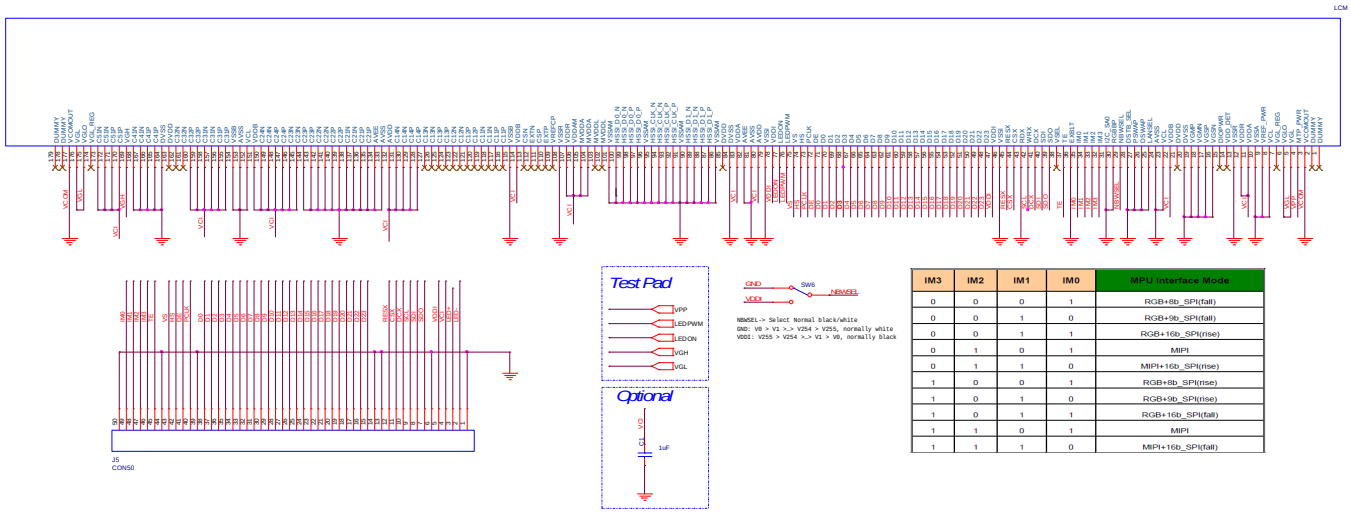
### 2.1.1 MIPI Interface




| DSWAP | PSWAP | Pins | | | | | |
|---|---|---|---|---|---|---|---|
| | | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| 0 | 0 | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| | 1 | CLK_N | CLK_P | D0_N | D0_P | D1_N | D1_P |
| 1 | 0 | CLK_P | CLK_N | D1_P | D1_N | D0_P | D0_N |
| | 1 | CLK_N | CLK_P | D1_N | D1_P | D0_N | D0_P |

| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(fall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(fall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(rise) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(fall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(fall) |

### 2.1.2 RGB Interface

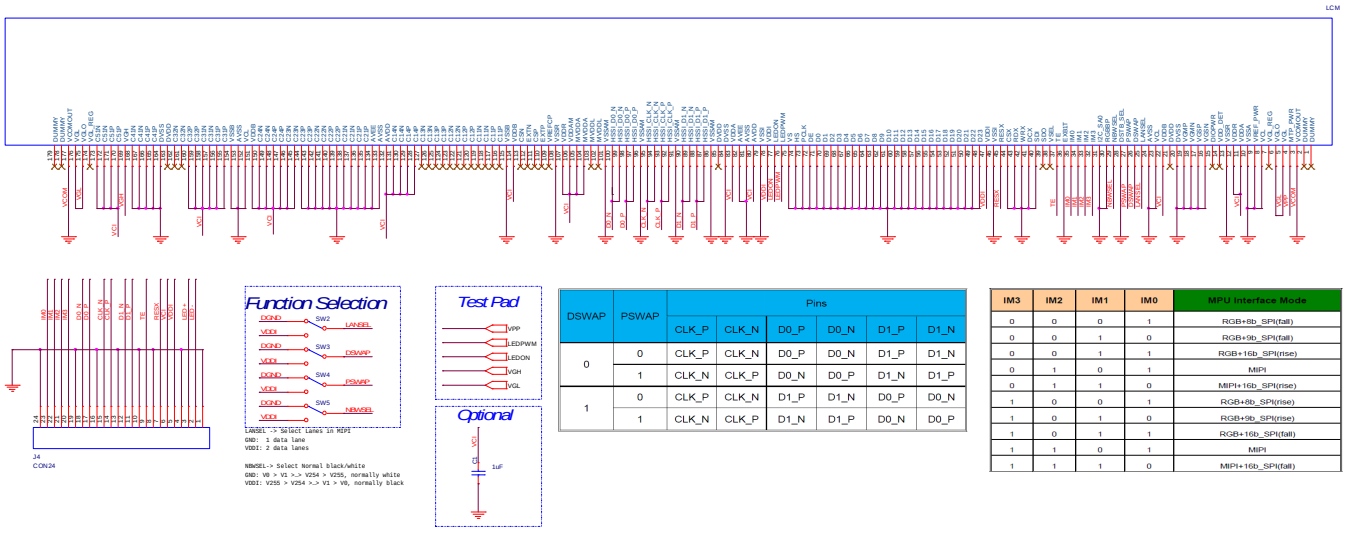



| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(fall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(fall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(rise) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(fall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(fall) |

## 2.2 IVO 4.5” TN (C045F8W7-C00)/ IPS (C045SWY2-0)

# Application Circuit

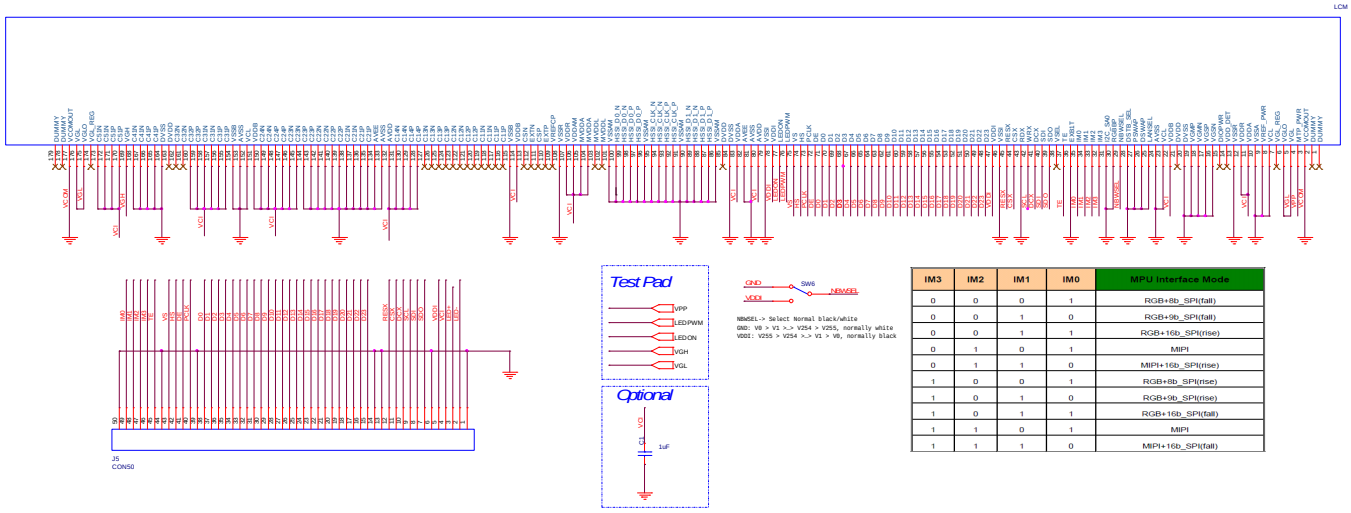
### 2.2.1 MIPI Interface




| DSWAP | PSWAP | Pins | | | | | |
|---|---|---|---|---|---|---|---|
| | | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| 0 | 0 | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| | 1 | CLK_N | CLK_P | D0_N | D0_P | D1_N | D1_P |
| 1 | 0 | CLK_P | CLK_N | D1_P | D1_N | D0_P | D0_N |
| | 1 | CLK_N | CLK_P | D1_N | D1_P | D0_N | D0_P |

| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(fall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(fall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(rise) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(fall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(fall) |

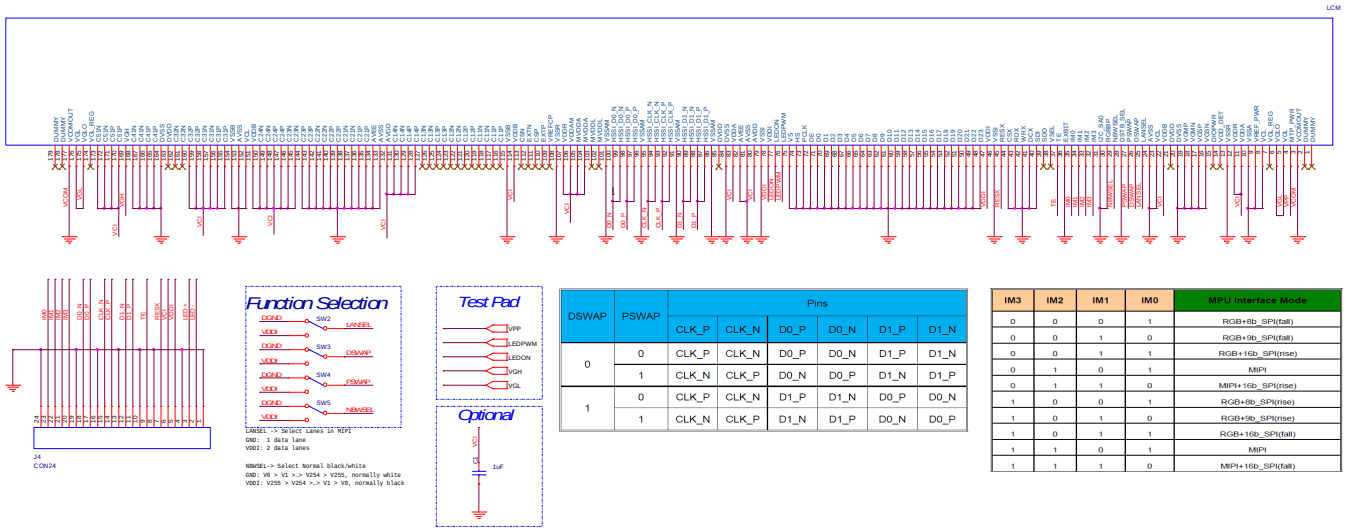
### 2.2.2 RGB Interface




| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(fall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(fall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(rise) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(fall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(fall) |

## 2.3 IVO 4.95 “(C050SWYG)/ 4.98” TN (C050SWY7-0) / IPS (C050SWY8-2) Application Circuit

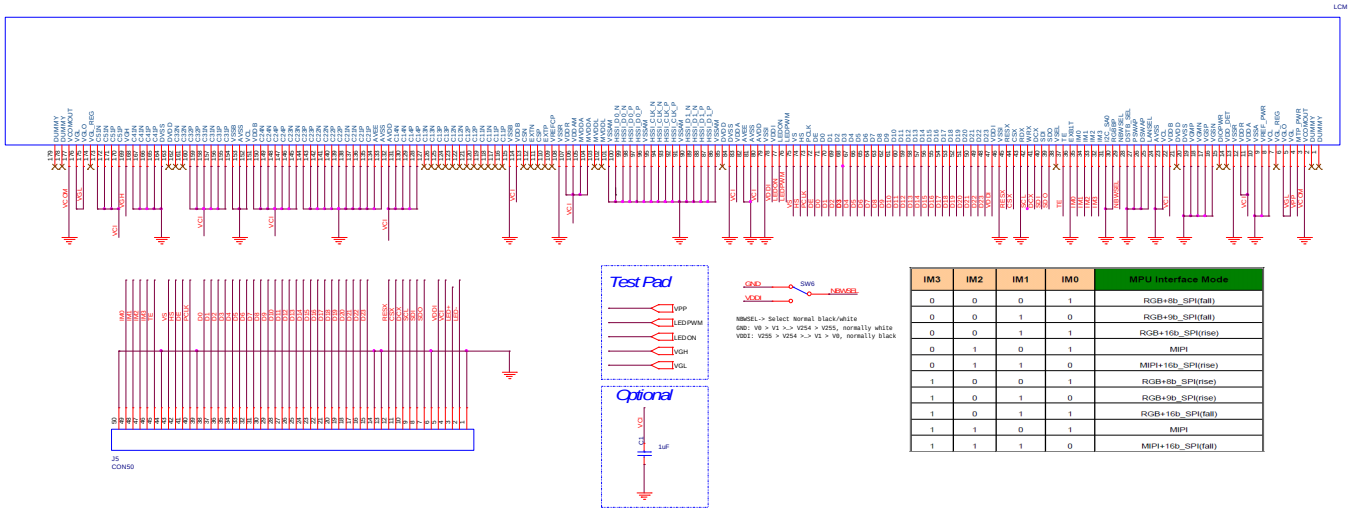
### 2.3.1 MIPI Interface




| DSWAP | PSWAP | Pins | | | | | |
|---|---|---|---|---|---|---|---|
| | | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| 0 | 0 | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| | 1 | CLK_N | CLK_P | D0_N | D0_P | D1_N | D1_P |
| 1 | 0 | CLK_P | CLK_N | D1_P | D1_N | D0_P | D0_N |
| | 1 | CLK_N | CLK_P | D1_N | D1_P | D0_N | D0_P |

| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(fall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(fall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(rise) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(fall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(fall) |

### 2.3.2 RGB Interface

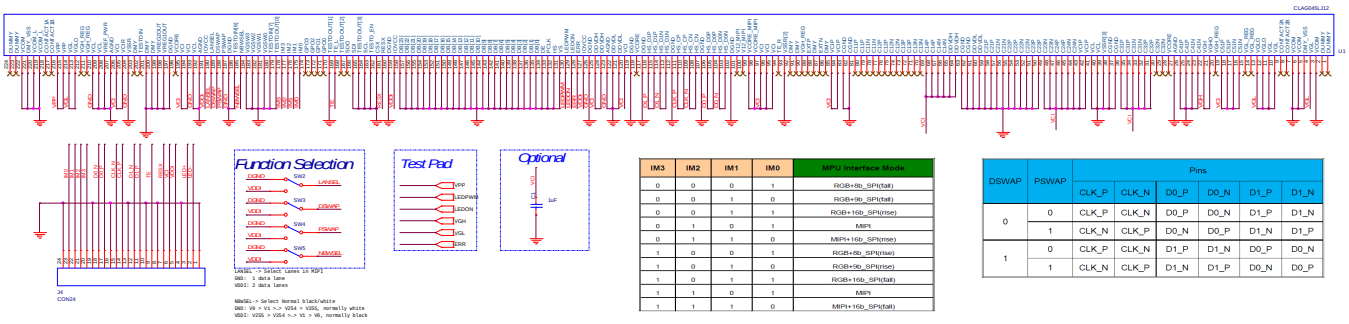



| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(fall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(fall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(rise) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(fall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(fall) |

# 3 CPT Panel

## 3.1 CPT 3.97(F040A16-604) Application Circuit

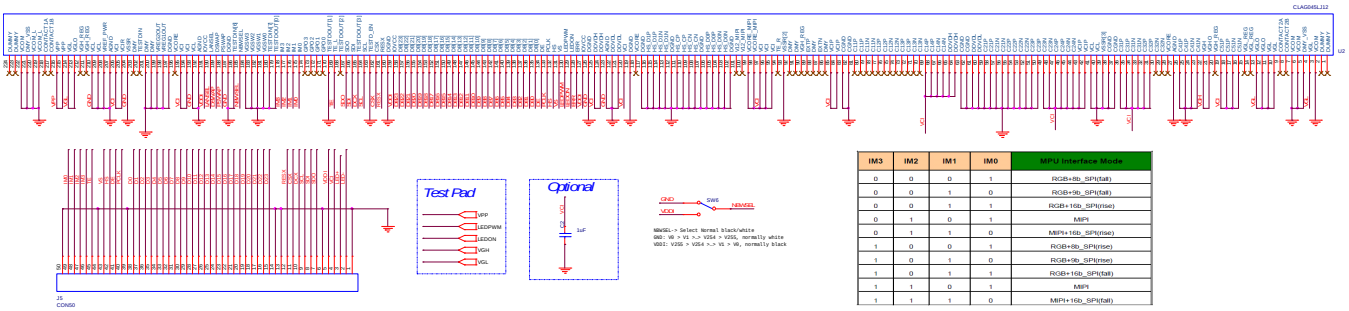
### 3.1.1 MIPI Interface




| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(fall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(fall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(rise) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(fall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(fall) |

| DSWAP | PSWAP | Pins | | | | | |
|---|---|---|---|---|---|---|---|
| | | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| 0 | 0 | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| | 1 | CLK_N | CLK_P | D0_N | D0_P | D1_N | D1_P |
| 1 | 0 | CLK_P | CLK_N | D1_P | D1_N | D0_P | D0_N |
| | 1 | CLK_N | CLK_P | D1_N | D1_P | D0_N | D0_P |

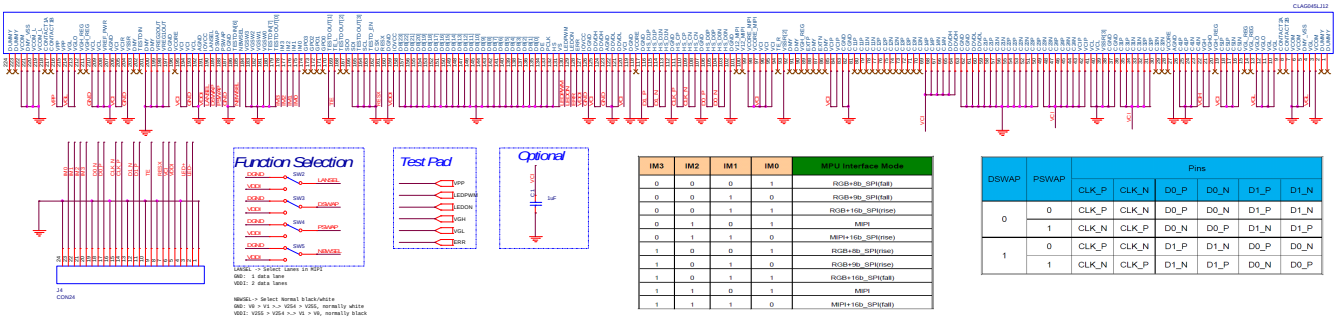
### 3.1.2 RGB Interface




| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(fall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(fall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(rise) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(fall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(fall) |

## 3.2 CPT 4.5(CLAG045LJ12) Application Circuit

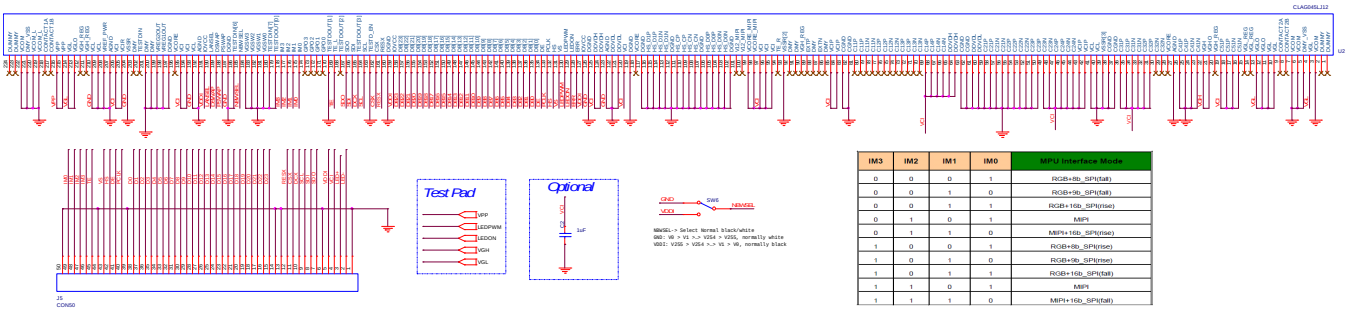
### 3.2.1 MIPI Interface




| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(fall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(fall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(rise) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(fall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(fall) |

| DSWAP | PSWAP | Pins | | | | | |
|---|---|---|---|---|---|---|---|
| | | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| 0 | 0 | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| | 1 | CLK_N | CLK_P | D0_N | D0_P | D1_N | D1_P |
| 1 | 0 | CLK_P | CLK_N | D1_P | D1_N | D0_P | D0_N |
| | 1 | CLK_N | CLK_P | D1_N | D1_P | D0_N | D0_P |

### 3.2.2 RGB Interface

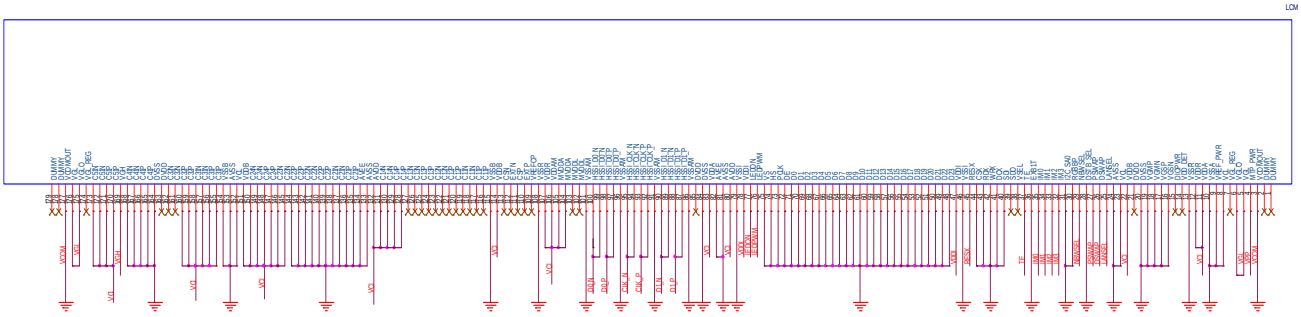


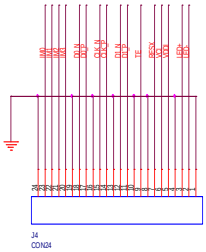

| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(fall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(fall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(rise) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(fall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(fall) |

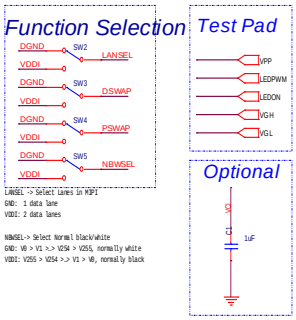
# 4 BOE Panel

## 4.1 BOE 3.97(B3 BD039WVS-810) Application Circuit

### 4.1.1 MIPI Interface







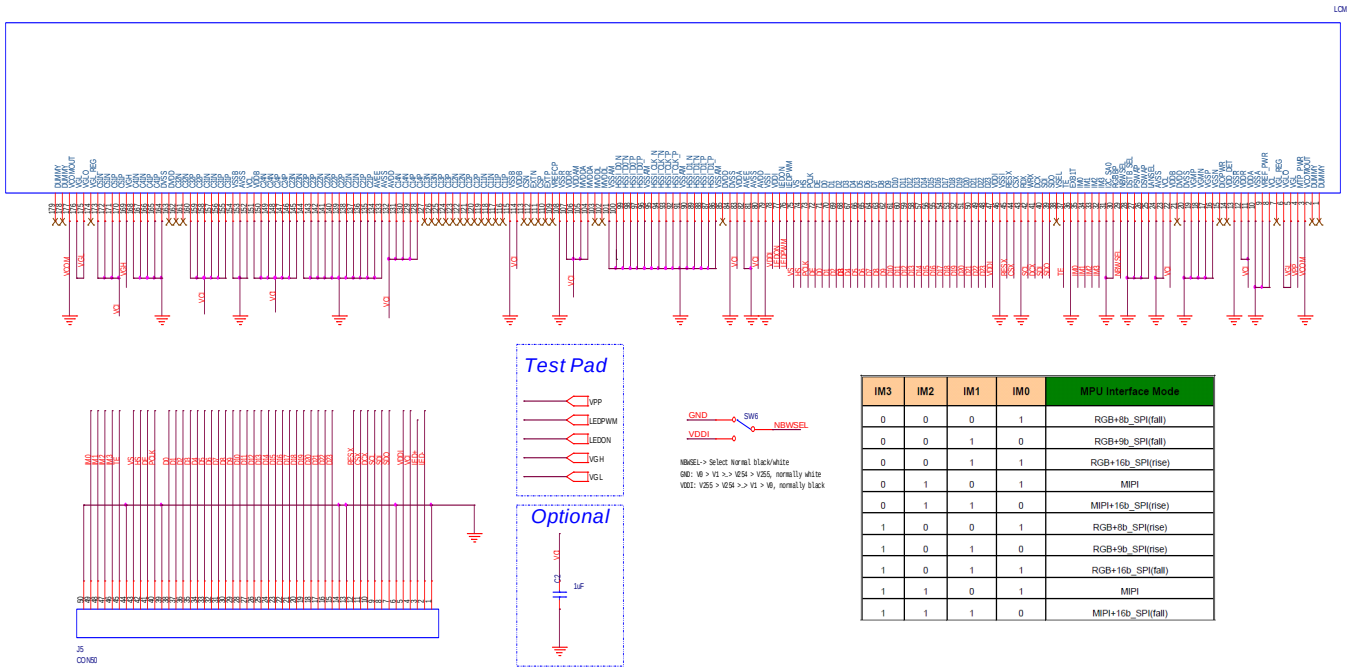




| DSWAP | PSWAP | Pins | | | | | |
|---|---|---|---|---|---|---|---|
| | | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| 0 | 0 | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| | 1 | CLK_N | CLK_P | D0_N | D0_P | D1_N | D1_P |
| 1 | 0 | CLK_P | CLK_N | D1_P | D1_N | D0_P | D0_N |
| | 1 | CLK_N | CLK_P | D1_N | D1_P | D0_N | D0_P |

| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(fall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(fall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(rise) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(fall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(fall) |

### 4.1.2 RGB Interface

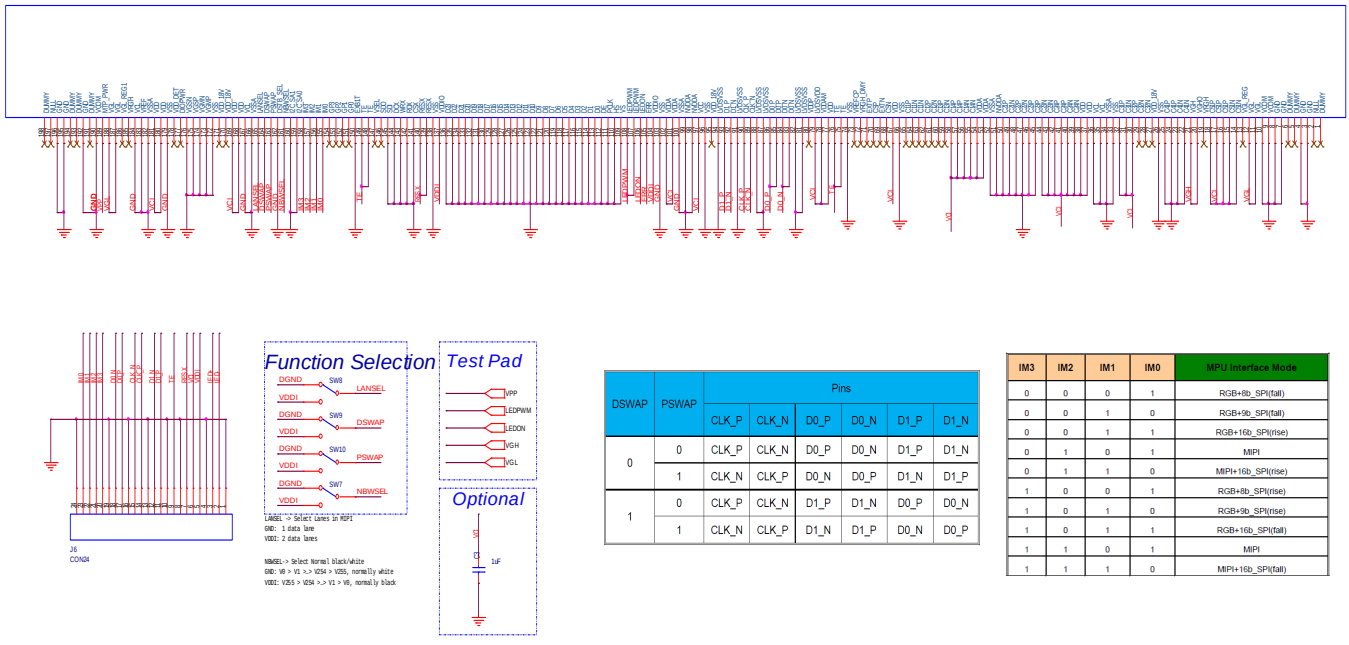



| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(fall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(fall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(rise) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(fall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(fall) |

## 4.2 BOE 4.5 TN (BT045FWQ-N80)/ IPS (BF045WVQ-100)

## Application Circuit

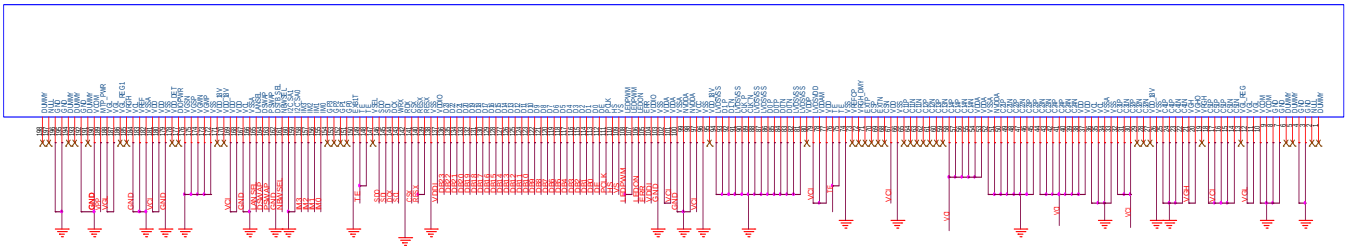
### 4.2.1 MIPI Interface

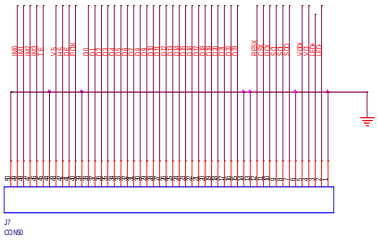


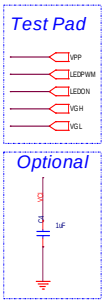

| DSWAP | PSWAP | Pins | | | | | |
|---|---|---|---|---|---|---|---|
| | | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| 0 | 0 | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| | 1 | CLK_N | CLK_P | D0_N | D0_P | D1_N | D1_P |
| 1 | 0 | CLK_P | CLK_N | D1_P | D1_N | D0_P | D0_N |
| | 1 | CLK_N | CLK_P | D1_N | D1_P | D0_N | D0_P |

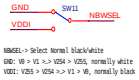
| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(fall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(fall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(rise) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(fall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(fall) |

### 4.2.2 RGB Interface







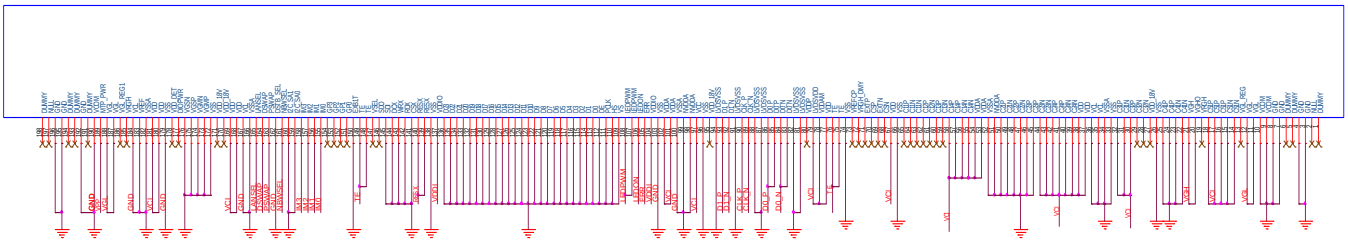



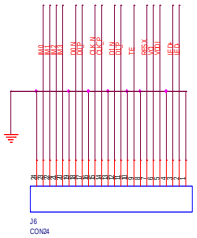

NBWSEL-> Select Normal black/white
GND: V0 > V1 >..> V254 > V255, normally white
VDDI: V255 > V254 >..> V1 > V0, normally black

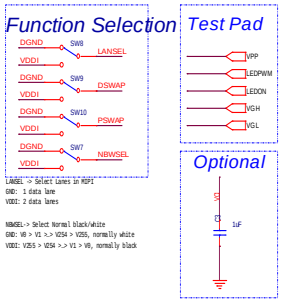
| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(fall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(fall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(rise) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(fall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(fall) |

## 4.3 BOE 5.0 TN (BT050FWQ-N80)/ IPS (BV050FWE-N43-1Q00) Application Circuit

### 4.3.1 MIPI Interface







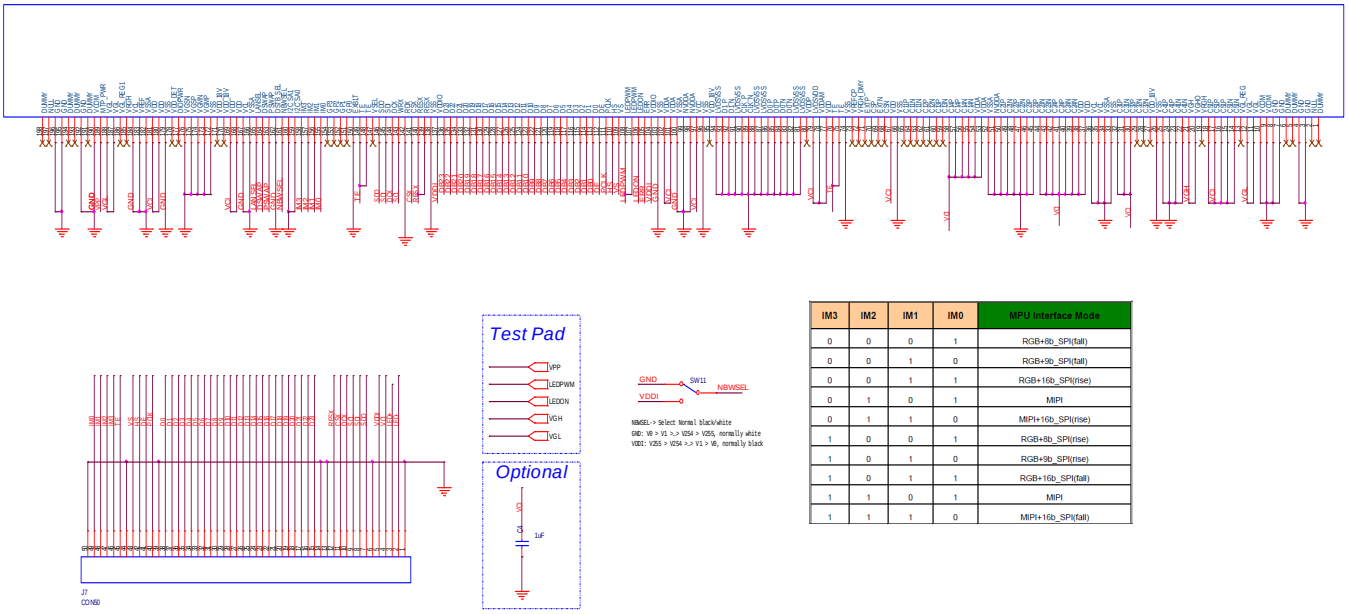




| DSWAP | PSWAP | Pins | | | | | |
|---|---|---|---|---|---|---|---|
| | | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| 0 | 0 | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| | 1 | CLK_N | CLK_P | D0_N | D0_P | D1_N | D1_P |
| 1 | 0 | CLK_P | CLK_N | D1_P | D1_N | D0_P | D0_N |
| | 1 | CLK_N | CLK_P | D1_N | D1_P | D0_N | D0_P |

| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(fall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(fall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(rise) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(fall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(fall) |

### 4.3.2 RGB Interface











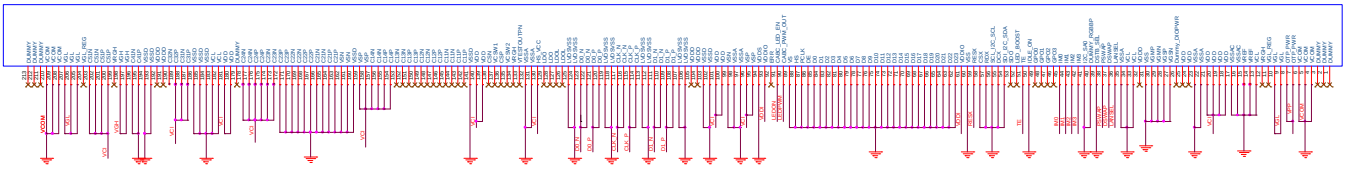
NBWSEL-> Select Normal black/white
GND: V0 > V1 >..> V254 > V255, normally white
VDDI: V255 > V254 >..> V1 > V0, normally black

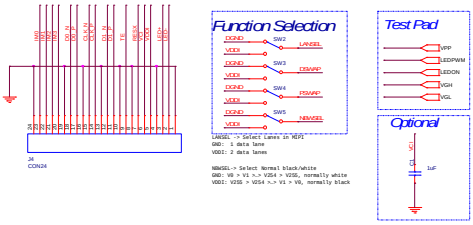
| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(fall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(fall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(rise) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(fall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(fall) |

# 5 CMI Panel

## 5.1 CMI 5.5 Application Circuit

### 5.1.1 MIPI Interface





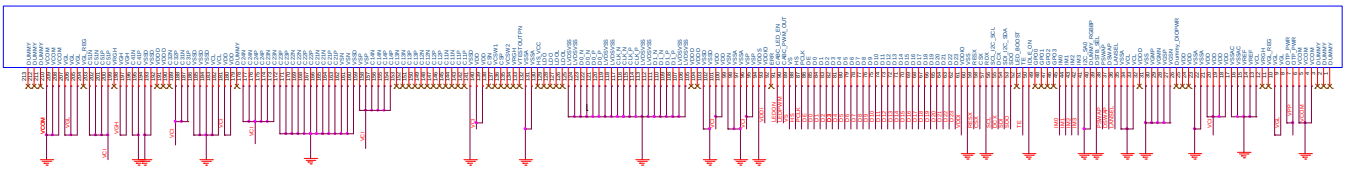

LANSEL-> Select Lanes in MIPI
GND: 1 data lane
VDDI: 2 data lanes

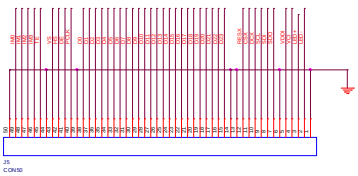
NBWSEL-> Select Normal black/white
GND: V0 > V1 >..> V254 > V255, normally white
VDDI: V255 > V254 >..> V1 > V0, normally black

| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(fall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(fall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(rise) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(fall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(fall) |

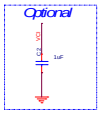
| DSWAP | PSWAP | Pins | | | | | |
|---|---|---|---|---|---|---|---|
| | | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| 0 | 0 | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| | 1 | CLK_N | CLK_P | D0_N | D0_P | D1_N | D1_P |
| 1 | 0 | CLK_P | CLK_N | D1_P | D1_N | D0_P | D0_N |
| | 1 | CLK_N | CLK_P | D1_N | D1_P | D0_N | D0_P |

### 5.1.2 RGB Interface









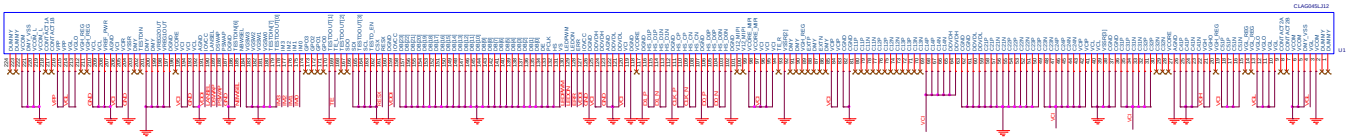


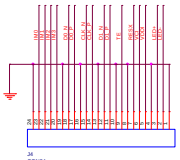
| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(fall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(fall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(rise) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(fall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(fall) |

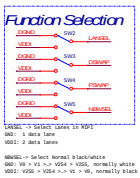
# 6 CTC Panel

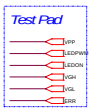
## 6.1 CTC 3.97 (PH040NA-01B) Application Circuit

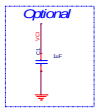
### 6.1.1 MIPI Interface







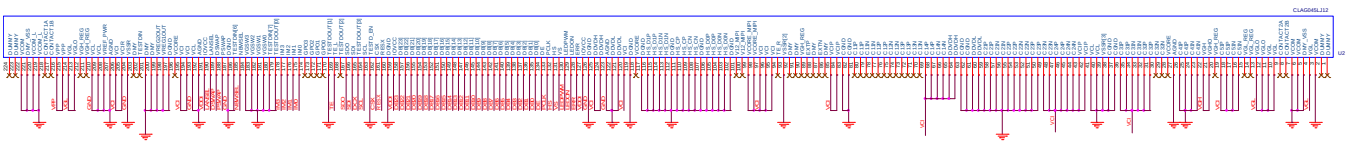



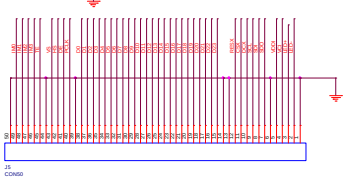



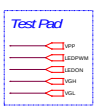

| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(fall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(fall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(rise) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(fall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(fall) |

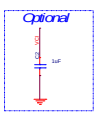
| DSWAP | PSWAP | Pins | | | | | |
|---|---|---|---|---|---|---|---|
| | | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| 0 | 0 | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| | 1 | CLK_N | CLK_P | D0_N | D0_P | D1_N | D1_P |
| 1 | 0 | CLK_P | CLK_N | D1_P | D1_N | D0_P | D0_N |
| | 1 | CLK_N | CLK_P | D1_P | D1_N | D0_N | D0_P |

### 6.1.2 RGB Interface







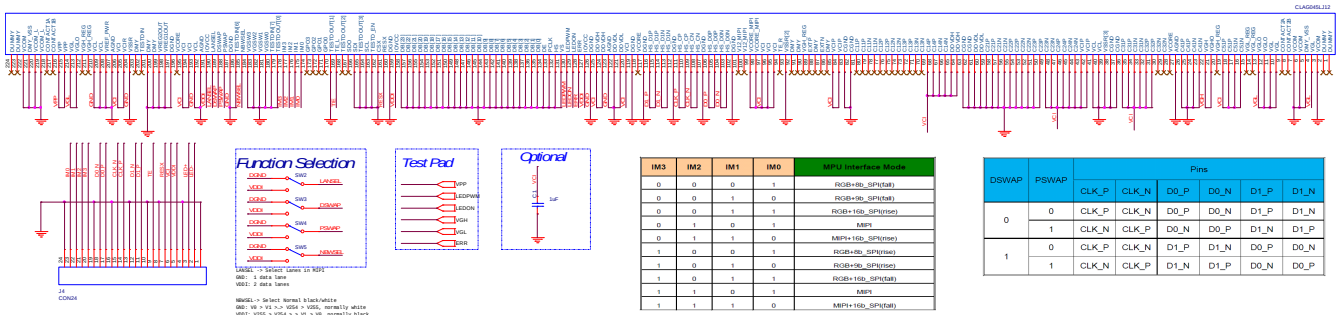




| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(fall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(fall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(rise) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(fall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(fall) |

## 6.2 CTC 4.5 () Application Circuit

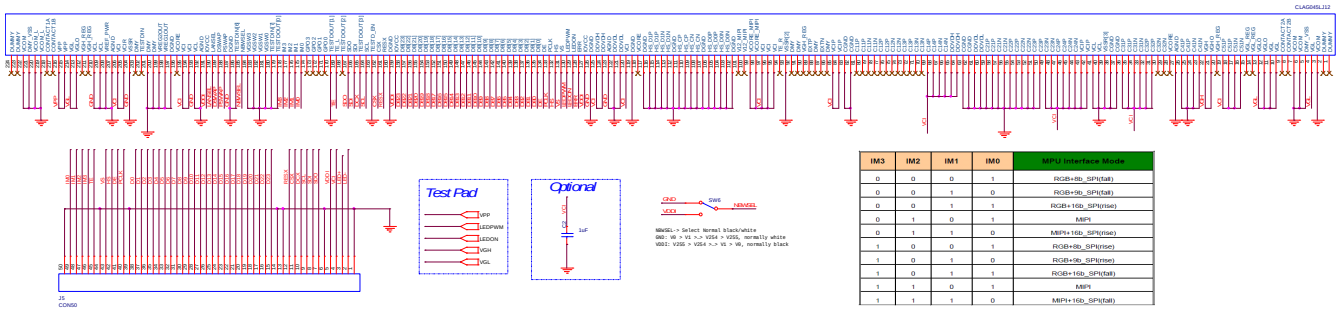
### 6.2.1 MIPI Interface




| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(tall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(tall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(rise) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(tall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(tall) |

| DSWAP | PSWAP | Pins | | | | | |
|---|---|---|---|---|---|---|---|
| | | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| 0 | 0 | CLK_P | CLK_N | D0_P | D0_N | D1_P | D1_N |
| | 1 | CLK_N | CLK_P | D0_N | D0_P | D1_N | D1_P |
| 1 | 0 | CLK_P | CLK_N | D1_P | D1_N | D0_P | D0_N |
| | 1 | CLK_N | CLK_P | D1_N | D1_P | D0_N | D0_P |

### 6.2.2 RGB Interface




| IM3 | IM2 | IM1 | IM0 | MPU Interface Mode |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | RGB+8b_SPI(tall) |
| 0 | 0 | 1 | 0 | RGB+9b_SPI(tall) |
| 0 | 0 | 1 | 1 | RGB+16b_SPI(rise) |
| 0 | 1 | 0 | 1 | MIPI |
| 0 | 1 | 1 | 0 | MIPI+16b_SPI(rise) |
| 1 | 0 | 0 | 1 | RGB+8b_SPI(rise) |
| 1 | 0 | 1 | 0 | RGB+9b_SPI(rise) |
| 1 | 0 | 1 | 1 | RGB+16b_SPI(tall) |
| 1 | 1 | 0 | 1 | MIPI |
| 1 | 1 | 1 | 0 | MIPI+16b_SPI(tall) |

# 7 Support Panel Cell Model

| No | Glass Company | Model Name | Size |
|---|---|---|---|
| | HSD | | 3.97” |
| | HSD | | 4.5” |
| | IVO | | 3.97” |
| | IVO | | 4.5” |
| | CPT | | 3.97” |
| | CPT | | 4.5” |
| | BOE | | 3.97” |
| | CTC | | 3.97” |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |